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On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

1.

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3. $H = I - 2uu^T$

a) $H^2 = HH = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - 2I(uu^T) - 2(uu^T)I + 4(uu^T)(uu^T)$

Now, $u^Tu = [a_1 a_2 \dots a_n] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

$= \sum_{i=1}^{n} a_i^2 = 1$

$H^2 = I^2 - 2I(uu^T) - 2(uu^T)I + 4u(u^Tu)u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot 1 \cdot u^T$ since, $u^Tu = 1$

$= I - 4uu^T + 4uu^T$

$H^2 = I$

b) Now from $H = H^T \rightarrow HH = HH^T$

$H^2 = HH^T$

$HH^T = I$ since $H^2 = I$

$H^TH = I$ so, $HH^T = H^TH = I$

c) Given that.

$H = I - 2uu^T$

$\Rightarrow Hu = (I - 2uu^T)u$ (multiplying by u in right side)

$\Rightarrow Hu = u - 2(uu^T)u$

$\Rightarrow Hu = u - 2u(u^Tu)$

$\Rightarrow Hu = u - 2u \cdot 1$ since, $u^Tu = 1$

$\Rightarrow Hu = u - 2u$

$\Rightarrow Hu = -u$

$Hu = (-1)u$

So one eigen vector of H is u itself

& its coresponding eigen value is $= -1$

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